

PF Lab 6 tasks (6)

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Task 5

```
#include<stdio.h>

int main()
{
    int num;

    double fact=1;

    printf("Enter a number: ");

    scanf("%d",&num);

    if(num<0){

        printf("Invalid number! Please enter a positive number\n");

    }

    else{

        for(int i=num ; i>=1 ; i--){

            fact *= i;

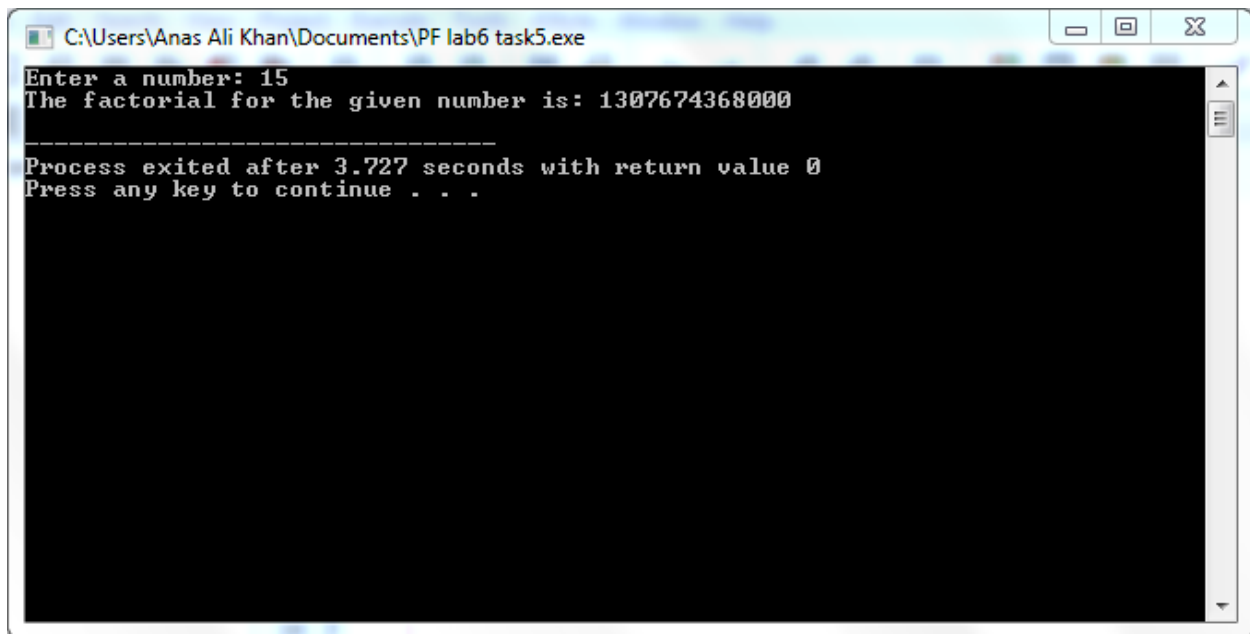
        }

    }

    printf("The factorial for the given number is: %.0lf\n",fact);

    return 0;

}
```



```
C:\Users\Anas Ali Khan\Documents\PF lab6 task5.exe
Enter a number: 15
The factorial for the given number is: 1307674368000
-----
Process exited after 3.727 seconds with return value 0
Press any key to continue . . .
```

Task 6

```
#include<stdio.h>

int main()
{
    int grade, sum=0, count=0;

    int a=0, b=0, c=0;

    float avg;

    printf("Enter your grade(0-100): ");

    do{

        scanf("%d",&grade);

        if(grade!=-1){

            printf("Your grade is %d\n",grade);

        }

        if(grade>70){
```

```
        a+=1;
    }
    else if(grade>50 && grade<70){
        b+=1;
    }
    else if(grade<50){
        c+=1;
    }

    sum+=grade;
    count+=1;
}while(grade!=-1);

avg=sum/count;

printf("Your average grade is:%.2f\n",avg);
printf("Number of grades above 70 are:%d\n",a);
printf("Number of grades between 50 and 70 are:%d\n",b);
printf("Number of grades below 50 are:%d\n",c);

return 0;

}
```

```
C:\Users\Anas Ali Khan\Documents\PF lab6 task6.exe
Enter your grade(0-100): 87
Your grade is 87
67
Your grade is 67
90
Your grade is 90
23
Your grade is 23
54
Your grade is 54
12
Your grade is 12
78
Your grade is 78
65
Your grade is 65
-1
Your average grade is:52.00
Number of grades above 70 are:3
Number of grades between 50 and 70 are:3
Number of grades below 50 are:3

-----
Process exited after 27.86 seconds with return value 0
Press any key to continue . . .
```

Task 7

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int balance=5000;
```

```
    double newbalance=5000;
```

```
    int deposit, withdrawal, amount;
```

```
    int w=0;
```

```
    int d=0;
```

```
    float final_bal;
```

```
    do{
```

```
        printf("Enter the amount you want to deposit/withdrawal(positive=deposit,  
negative=withdrawal): ");
```

```
        scanf(" %d",&amount);
```

```
        if(amount>0){
```

```
        deposit=amount;

        newbalance=newbalance+deposit;

        d+=1;
    }

    else if(amount<0){

        withdrawal=amount;

        newbalance=newbalance+withdrawal;

        w+=1;
    }


    printf("\nYour updated balance is:$ %.2lf\n",newbalance);


}while(amount!=0);


    printf("The number of deposits are: %d\n",d);

    printf("The number of withdrawals are: %d\n",w);

    final_bal=newbalance;

    printf("\nYour final balance is: %.2lf\n",final_bal);


    return 0;

}
```

```
C:\Users\Anas Ali Khan\Documents\PF lab6 task7.exe
Enter the amount you want to deposit/withdrawal(positive=deposit, negative=withd
rawal): 2000
Your updated balance is:$ 7000.00
Enter the amount you want to deposit/withdrawal(positive=deposit, negative=withd
rawal): 1500
Your updated balance is:$ 8500.00
Enter the amount you want to deposit/withdrawal(positive=deposit, negative=withd
rawal): -3000
Your updated balance is:$ 5500.00
Enter the amount you want to deposit/withdrawal(positive=deposit, negative=withd
rawal): -2500
Your updated balance is:$ 3000.00
Enter the amount you want to deposit/withdrawal(positive=deposit, negative=withd
rawal): 4500
Your updated balance is:$ 7500.00
Enter the amount you want to deposit/withdrawal(positive=deposit, negative=withd
rawal): 0
Your updated balance is:$ 7500.00
The number of deposits are: 3
The number of withdrawals are: 2
Your final balance is: 7500.00
-----
Process exited after 56.03 seconds with return value 0
Press any key to continue . . .
```

Task 8

```
#include<stdio.h>

int main()
{
    int num;

    printf("Enter the number you want the table for: ");

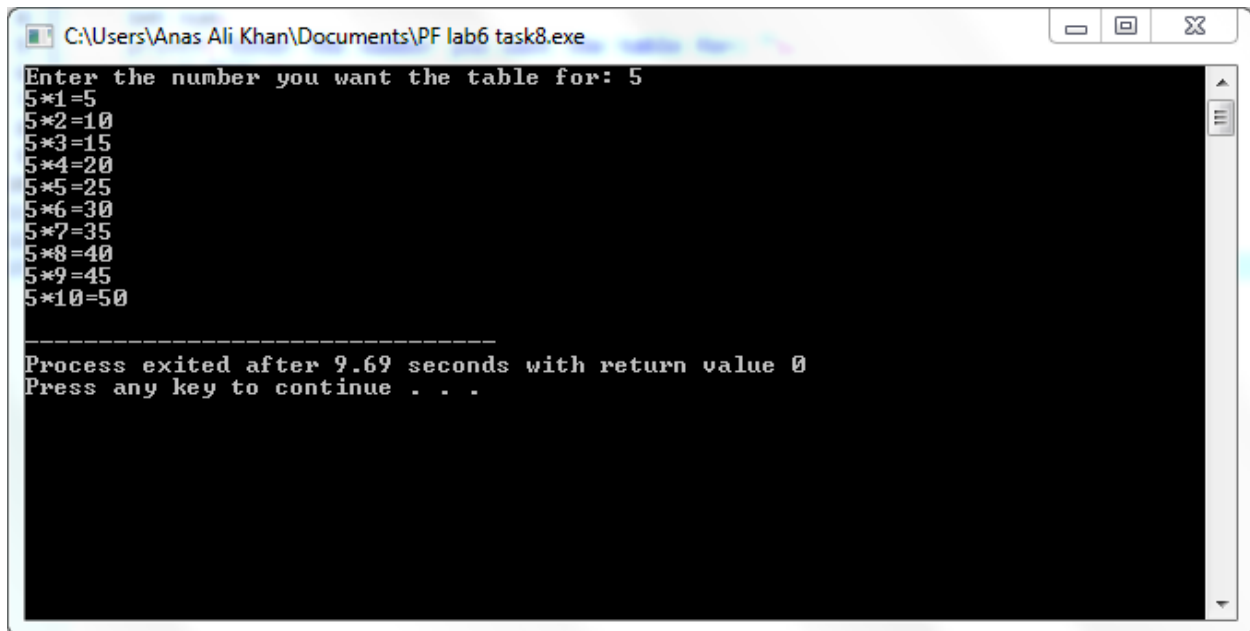
    scanf("%d",&num);

    for(int i=1;i<=10;i++){

        printf("%d*%d=%d\n",num,i,num*i);

    }

    return 0;
}
```



```
C:\Users\Anas Ali Khan\Documents\PF lab6 task8.exe
Enter the number you want the table for: 5
5*1=5
5*2=10
5*3=15
5*4=20
5*5=25
5*6=30
5*7=35
5*8=40
5*9=45
5*10=50
-----
Process exited after 9.69 seconds with return value 0
Press any key to continue . . .
```


Task 9

```
#include<stdio.h>

int main()
{
    int num;

    int largest, smallest;

    char decision;

    do{
        printf("Enter a number: ");

        scanf(" %d",&num);

        if(num>largest){
            largest=num;

            printf("The largest number is: %d\n",largest);
        }

        else if(num<smallest){
            smallest=num;

            printf("The smallest number entered is: %d\n",smallest);
        }

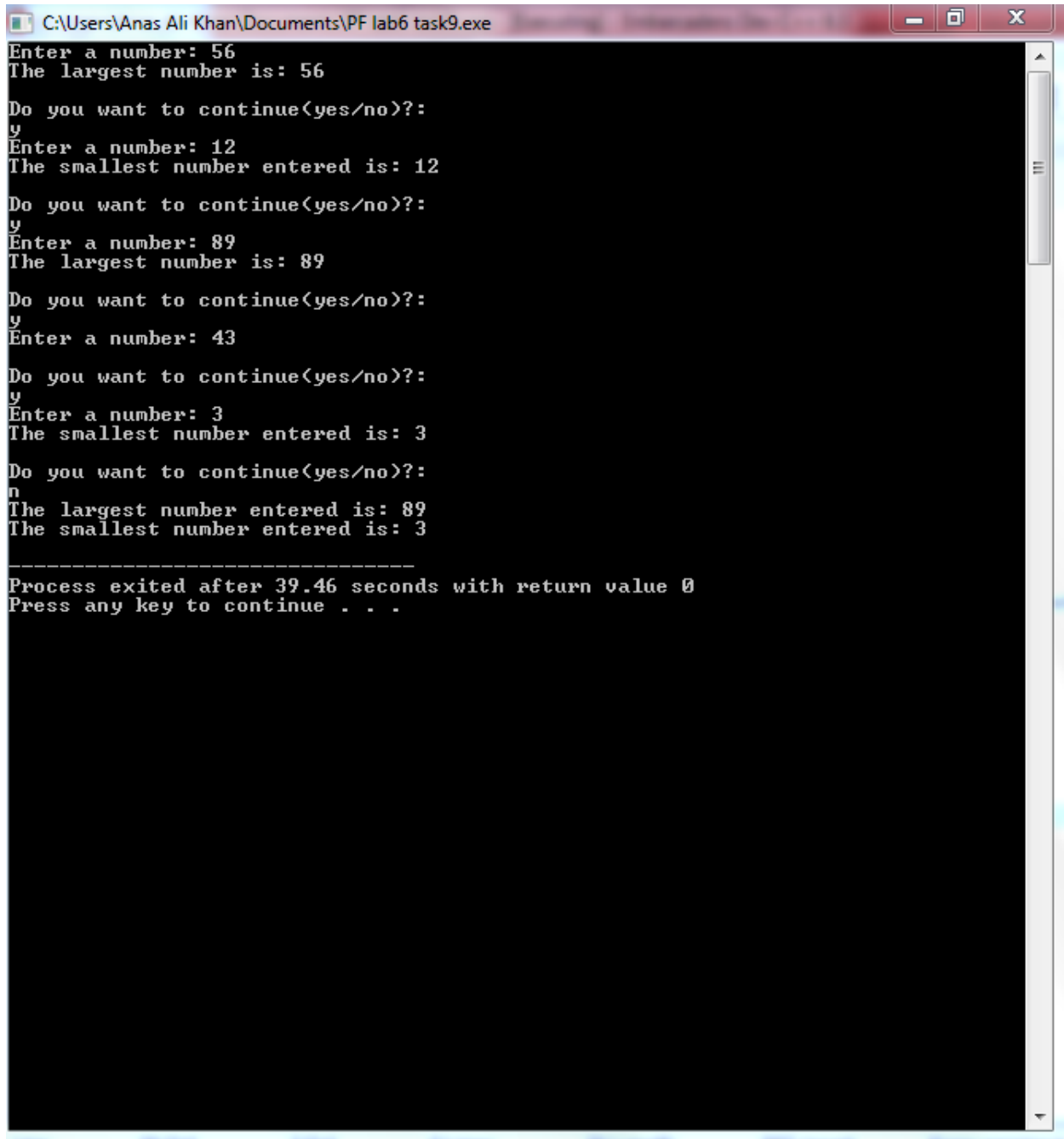
        printf("\nDo you want to continue(yes/no)? : \n");

        scanf(" %c",&decision);

    }while(decision!='n');

    printf("The largest number entered is: %d\n",largest);
```

```
printf("The smallest number entered is: %d\n",smallest);  
}
```



```
C:\Users\Anas Ali Khan\Documents\PF lab6 task9.exe  
Enter a number: 56  
The largest number is: 56  
Do you want to continue(yes/no)?:  
y  
Enter a number: 12  
The smallest number entered is: 12  
Do you want to continue(yes/no)?:  
y  
Enter a number: 89  
The largest number is: 89  
Do you want to continue(yes/no)?:  
y  
Enter a number: 43  
Do you want to continue(yes/no)?:  
y  
Enter a number: 3  
The smallest number entered is: 3  
Do you want to continue(yes/no)?:  
n  
The largest number entered is: 89  
The smallest number entered is: 3  
-----  
Process exited after 39.46 seconds with return value 0  
Press any key to continue . . .
```

Task 10

```
#include<stdio.h>

int main()
{
    int num, total=0;

    char decision;

    while(decision='y'){

        printf("Enter a number: ");

        scanf("%d",&num);

        total+=num;

        printf("The running total is: %d\n",total);

        printf("Do you want to continue(y/n)? : ");

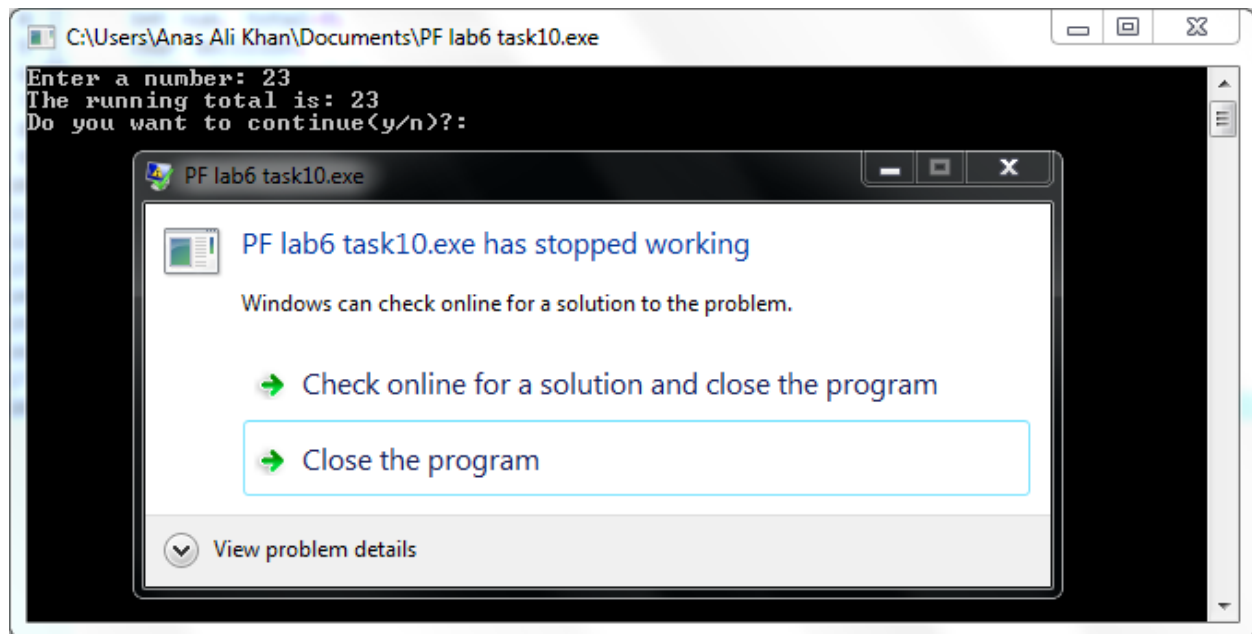
        scanf("%c",decision);

    }

    printf("\nThe total is: %d",total);

    return 0;

}
```



The feels keep crashing everytime I run it.